

MOHAMMED SAHAL SHAIKH

Mechanical Engineering Graduate · CFD & Simulation · Research · Sustainable Design

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South African Permanent Resident • ECSA Candidate Engineer registration in progress (SAQA evaluation underway)

PROFESSIONAL SUMMARY

Mechanical Engineering (Honours) graduate from Curtin University (2026) with a 72.89% cumulative average and four peer-reviewed publications. Combines hands-on CAD, simulation (ANSYS, MATLAB, Simulink), and materials-characterisation skills, demonstrated in a CFD-validated Honours thesis on natural oil sorbents. Founding leadership of multiple student engineering bodies and two first-place national competition wins. Seeking a graduate mechanical, research, or design engineering role, translating simulation and materials insight into measurable, sustainable outcomes.

TECHNICAL SKILLS

CAD & Design: SolidWorks, Autodesk Inventor Professional, AutoCAD, FreeCAD, GD&T certification (ASME)
Simulation & Analysis: ANSYS (Fluent / Mechanical), MATLAB, Simulink, SimScale: CFD & Finite Element Analysis (FEA)
Programming & Data: Python (OpenCV, image analysis), Arduino IDE, Advanced Microsoft Excel
Manufacturing & Prototyping: Sheet-metal fabrication, soldering, 3D printing, laser cutting
Electronics (EDA): LTspice, Cadence
Languages: English, Afrikaans, Urdu (fluent); Turkish (conversational)

EDUCATION

BEng (Honours), Mechanical Engineering — Curtin University, Dubai 2022 – 2026
Cumulative Weighted Average: 72.89% • Final-Year Average: 78%
Relevant coursework: Computational Fluid Dynamics and Finite Element Analysis (both scored above 80%), Thermodynamics, Heat Transfer, Engineering Management.
Honours Thesis: “Investigating the Structure of *Solanum melongena* L. for Sustainable Oil Absorption and Environmental Remediation.” *Full technical detail under Selected Projects below.*

PROFESSIONAL EXPERIENCE

Solar Energy Research Intern — Transnational Academic Group & Curtin University Feb 2026 – Jun 2026
– Designed, built and tested an instrumented solar-panel soiling rig to quantify how dust reduces panel efficiency and service life, producing the dataset used to inform the site's cleaning and maintenance schedule.
– Synthesised 20+ peer-reviewed articles into a journal-standard, publication-ready report on panel degradation.

STEM / CAD Instructor — The Logix Engine Oct 2025 – Feb 2026
– Coached a First Lego League team end-to-end through the robot and innovation challenges, applied soldering, wiring, 3D printing and laser cutting skills.
– Delivered CAD, robotics and STEM workshops across multiple schools and built new CAD courses and project curricula for senior students.

Project Coordinator — ICISAS 2025 — Transnational Academic Group Dec 2024 – Mar 2025
– Screened and triaged 450+ research submissions for an international conference, enforcing scope and quality standards.
– Directed peer review across 1,000+ reviewers, running the end-to-end submission system and reviewer assignments.

Materials & Solar Research Intern — Transnational Academic Group & Curtin University Aug 2023 – Mar 2024
– Benchmarked 5 solar-panel technologies, mapping the optimal applications and trade-offs of each.
– Evaluated 2 aluminium-alloy combinations and the effect of manufacturing processes on their mechanical properties.

Aviation Maintenance Intern — Sarsan Aviation Academy (GCAA CAR-147) Aug 2023
– Completed hands-on airframe maintenance training on the Piper PA-28 — sheet-metal repair, safety wiring, wing attachment and flight-control rigging.
– Completed GCAA-approved training in aerodynamics, gas-turbine engines, avionics and aviation regulations.

SELECTED PROJECTS

Bio-Inspired Oil Sorbent Characterisation — Honours Thesis

Developed a validated three-phase framework characterising eggplant (*Solanum melongena* L.) mesocarp as a natural oil sorbent, transferable to any agricultural biosorbent material.

- Built a Python / OpenCV image-analysis pipeline (Otsu thresholding + connected-component analysis) across 59 cross-sections, detecting 4,498 pores and deriving porosity (22.2%), mean pore diameter (954 μm) and permeability ($9.16 \times 10^{-11} \text{ m}^2$).
- Constructed a 3D porous-zone CFD model in ANSYS Fluent (Darcy–Forchheimer, laminar regime) that predicted permeability within 0.17% of the image-derived value, cross-validating the structural characterisation.
- Ran a gravimetric absorption protocol on 26 specimens across 4 fluids; dehydrated samples reached 1.69 g/g — 96.6% of the theoretical maximum uptake, quantifying the material’s remediation potential.

Solar Cooking Backpack — James Dyson Award, National Finalists (2023)

- Led the team design of a portable solar cooker for off-grid communities; developed the thermal-collector geometry in SolidWorks and prototyped a unit that cooked a full meal in ~2 hours, roughly 65% faster than benchmark solar cookers.

Current Research (under review for publication in ICISAS 2026)

- **Thassos Marble–Aluminium Composites:** analysing the mechanical and thermal properties of a marble–aluminium composite for passive cooling in data centres and AI server rooms.
- **CNT-Enhanced Sustainable Aviation Fuel:** technological, economic and lifecycle (TEA / LCA) scenario analysis of carbon-nanotube-enhanced synthetic SAF pathways.
- **AI-Driven Aviation Decarbonisation:** scenario-intelligence model built on the Boeing Cascade Climate Impact Model with a machine-learning surrogate layer.

LEADERSHIP & PROFESSIONAL INVOLVEMENT

President, Student Club — Project Management Institute (PMI), UAE Chapter

Aug 2025 – Dec 2026

- Spearheaded a governance restructure, added a Treasurer role, passed two formal bylaw amendments, and directed committee officers and the events programme.
- Organised a careers-readiness event for 40–50 attendees, owning budget, agenda, speaker outreach and logistics.

Founding President — IEEE Student Branch, Curtin University Dubai (Jan 2025 – Jun 2026): Chaired the ASHRAE student branch, planning technical workshops and guest-speaker sessions.

President — ASHRAE Student Branch, Curtin University Dubai (Jul 2025 – Jun 2026): established and chaired the branch, planning technical workshops and guest-speaker sessions.

Treasurer — Engineers Australia, Curtin University Dubai (Feb 2023 – Jun 2026): oversaw event budgets and finances across a three-year tenure, funding technical and professional-development activities.

PUBLICATIONS

Core mechanical / materials publication:

- Gadher, S., Shaikh, M.S.M.S., Selvan, C.P. (2025). *Effects on the Microstructure and Mechanical Properties of the AlSi14 Alloy Produced by the Selective Laser Melting (SLM) Technique*. ICISAS 2023, Springer Nature. doi:10.1007/978-3-031-68952-9_15

Additional peer-reviewed publications (interdisciplinary research — signal processing, medical data, cloud security):

- Rakesh, C., et al., Shaikh, M.S.M.S., Mahesh, N. *Self-Convolution Filter-Based ECG R-Peak Detection for Non-Stationary Signal Processing Applications*. IEEE ASET 2025. doi:10.1109/ASET66891.2025.11427994
- Borah, A.R., et al., Shaikh, M.S.M.S., Mahesh, N. *Enhanced Medical Data Diagnosis Using an Optimized Soft Computing Model and Feature Selection Algorithms*. IEEE ASET 2025. doi:10.1109/ASET66891.2025.11428087
- Borah, A.R., et al., Shaikh, M.S.M.S., Selvan, C.P. *Ensuring Data Integrity and Secured Storage in Cloud with MDFA-Based Encrypted Data Search and Keystamp Generation*. IEEE ASET 2025. doi:10.1109/ASET66891.2025.11427882

AWARDS & HONOURS

- **1st Place** — Mubarmij50 Robotics Competition (2024)
- **1st Place** — Manipal University Hydro Jet Competition (2022)
- **1st Runner-Up** — Western Australia Climate Action Challenge (2024)